

CODE SPORT



In this month's column, we discuss a new text mining task called machine reading comprehension.

Over the past couple of months, we have been discussing the problem of detecting duplicate questions in community question answering (CQA) forums using the Quora question pair data set. In this month's column, let's discuss a closely related problem, namely, machine reading comprehension. This is an important language-related task in artificial intelligence (AI).

Artificial intelligence has seen rapid progress over the last few years, with a number of state-of-art advances in vision, speech and language. Computing intelligence has evolved from simple number crunching tasks to perceptual computing tasks, whereby machines can take sensory inputs from the surrounding environment such as vision, speech, etc. The next step in the evolution of computing intelligence is cognitive intelligence computing, where machines can read language/text, understand meanings, reason based on the information they have read, and make decisions even in complex scenarios, on par with human intelligence. This requires that machines have the ability to read text, understand it, extract meaning from it and reason, based on its understanding.

There are a number of language related tasks pertaining to AI. These include information extraction, information search, dialogue, and of course question answering (QA). It is possible to frame all the other tasks as QA. For instance, information search can be framed as asking questions/queries, with the machines retrieving the relevant answer in an interactive fashion from the Web/ external knowledge bases. Dialogue can be framed as conversational QA, of which a popular example is Apple's Siri. Question answering is an AI complete problem, in the sense that finding a good solution to QA will enable solutions to other language related AI tasks.

QA became a popular AI task when IBM demonstrated it in the AI Grand Challenge. In the popular TV game show 'Jeopardy', the machine intelligence of IBM Watson was pitted against human contestants Ken Jennings and Brad

Ritter. IBM Watson used many of the natural language processing techniques such as 'similar question retrieval', summarisation, key passage identification and answer sentence extraction to beat the humans. This was possibly the first well-known demonstration of machines performing the QA task and fuelled the public's imagination.

Machine QA has a number of flavours. Given a question, the machine can look through external knowledge bases and try to find an answer. For this problem, it needs to convert the natural language question into a logical form, which can be used to query the external knowledge base. A particular NLP technique known as semantic parsing is used to convert a natural language query into a logical form, specific to the schema of the external knowledge base. The logical query can be used to retrieve results from the external knowledge base. While this can be used for structured external knowledge bases, there are a number of unstructured text documents that contain the information required to answer the questions. For instance, many of the factual questions can be answered using Wikipedia, which contains a number of articles on different topics. Hence, answering questions related to each of these topics requires reading the Wikipedia text. The machines can then answer questions that are based on the text read. This has given rise to a specific form of AI QA task known as machine reading comprehension (MRC).

Here is an MRC task: Given a passage P, and a question Q related to that passage, the QA system needs to provide answer A to the given question. There can be more than one question, as is typically the case in reading comprehension tasks. There can be two variants to this task. The simplest variant is the answer extraction task, where the exact answer A to question Q is present in the passage P. This is known as the answer extraction task. A more complicated variant is the answer generation task. In this case, the exact answer is not present directly in the passage P. Instead, the program has to synthesise/generate